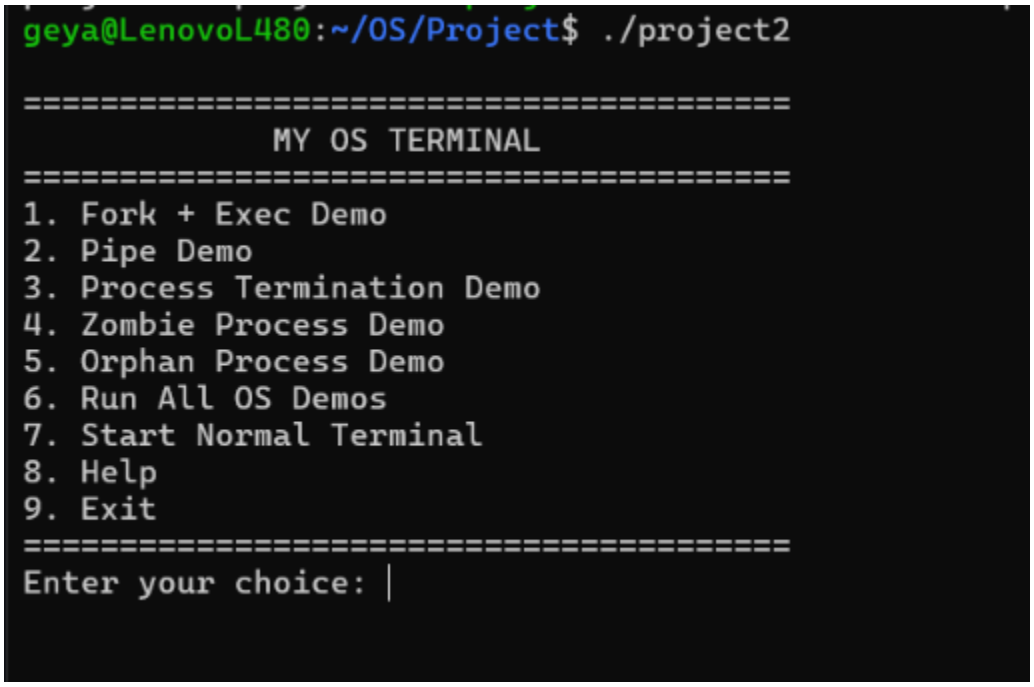
Menu:



Choice 1:

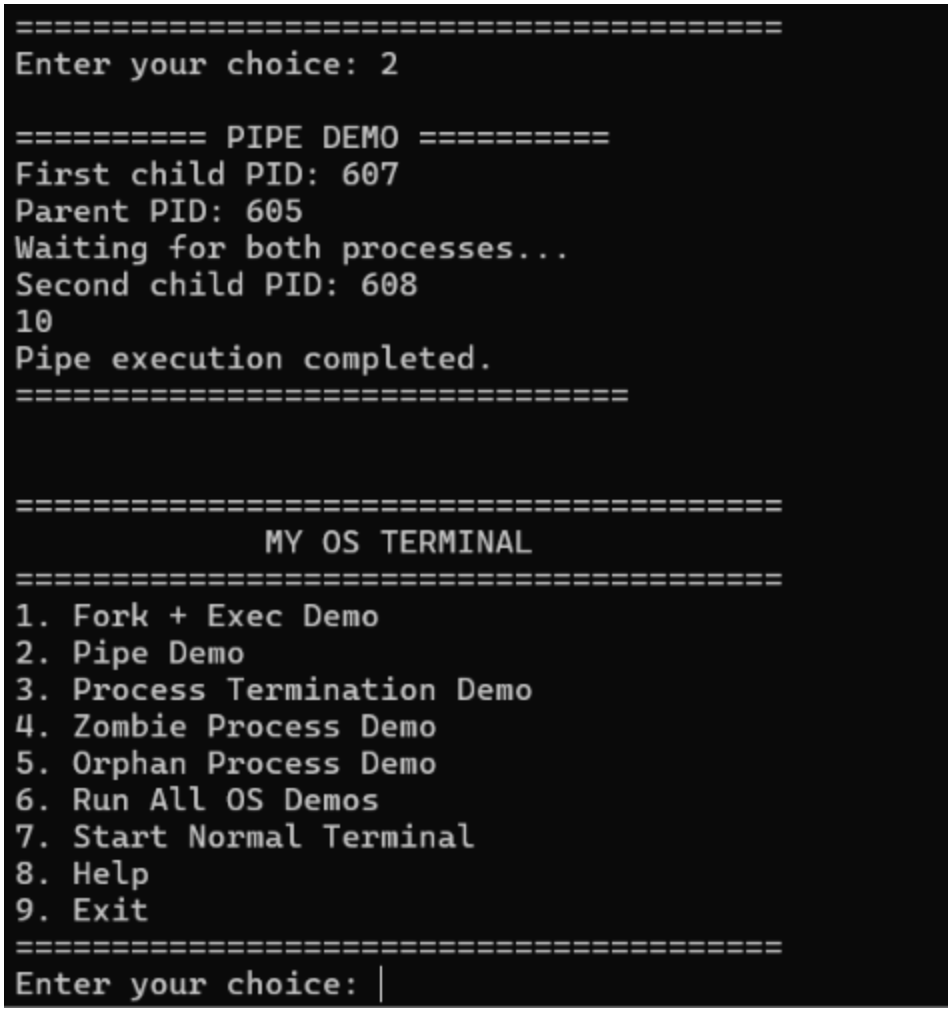
```
========================================
Enter your choice: 1

========== FORK + EXEC DEMO ==========
Parent PID : 605
Child PID  : 606
Parent is waiting for child...
Child process created
Child PID  : 606
Child PPID : 605
Child is now executing 'ls -l'

total 184
-rwxr-xr-x 1 geya geya 25904 Sep 22 06:57 project
-rw-r--r-- 1 geya geya 18296 Sep 24 10:00 project.c
-rwxr-xr-x 1 geya geya 25904 Oct  1 10:02 project1
-rw-r--r-- 1 geya geya 18324 Oct  1 10:02 project1.c
-rw-r--r-- 1 geya geya    25 Oct  1 10:02 project1.c:Zone.Identifier
-rwxr-xr-x 1 geya geya 25944 Oct  1 10:06 project2
-rw-r--r-- 1 geya geya 18820 Oct  1 10:06 project2.c
-rw-r--r-- 1 geya geya    25 Oct  1 10:06 project2.c:Zone.Identifier
-rwxr-xr-x 1 geya geya 25976 Oct  1 09:58 project_1
-rw-r--r-- 1 geya geya    25 Oct  1 09:57 project_1.c:Zone.Identifier
Child finished.
========================================


========================================
            MY OS TERMINAL
```

Choice 2:


==========
Enter your choice: 2
========== PIPE DEMO ==========
First child PID: 607
Parent PID: 605
Waiting for both processes...
Second child PID: 608
10
Pipe execution completed.
==========
==========
MY OS TERMINAL
==========
1. Fork + Exec Demo
2. Pipe Demo
3. Process Termination Demo
4. Zombie Process Demo
5. Orphan Process Demo
6. Run All OS Demos
7. Start Normal Terminal
8. Help
9. Exit
==========
Enter your choice:

Choice 3:

```
Enter your choice: 3

========== TERMINATION DEMO ==========
Parent PID: 605
Waiting for child...
Child PID: 609
Child is terminating using exit(5)
Child terminated normally.
Child exit status: 5
=======================================


=======================================
             MY OS TERMINAL
=======================================
1. Fork + Exec Demo
2. Pipe Demo
3. Process Termination Demo
4. Zombie Process Demo
5. Orphan Process Demo
6. Run All OS Demos
7. Start Normal Terminal
8. Help
9. Exit
=======================================
Enter your choice:
```

Choice 4:

```
========================================
Enter your choice: 4

========== ZOMBIE PROCESS DEMO ==========
Parent PID: 605
Child PID : 610

Child has terminated.
Parent has NOT called wait() yet.
Checking child state from /proc...
Child PID: 610
Child is exiting now...

Child process state:
State:  Z (zombie)

Now parent calls waitpid()...
Zombie removed.
========================================
```

Choice 5:

```
========================================
Enter your choice: 5

========== ORPHAN PROCESS DEMO ==========
Terminal PID          : 605
Demo parent PID       : 611
Demo parent PID       : 611
Created child PID     : 612
Orphan candidate PID : 612
Original PPID         : 611
Demo parent is now terminating...

Demo parent terminated.
Waiting for orphan demonstration...

Demo parent has terminated.
New PPID of orphan    : 391
This process is now an orphan.
Orphan demonstration completed.
========================================
```

Choice 6:

```
##########################################
#       OS PROCESS DEMONSTRATIONS        #
##########################################

========== FORK + EXEC DEMO ==========
Parent PID : 605
Child PID  : 613
Parent is waiting for child...
Child process created
Child PID  : 613
Child PPID : 605
Child is now executing 'ls -l'

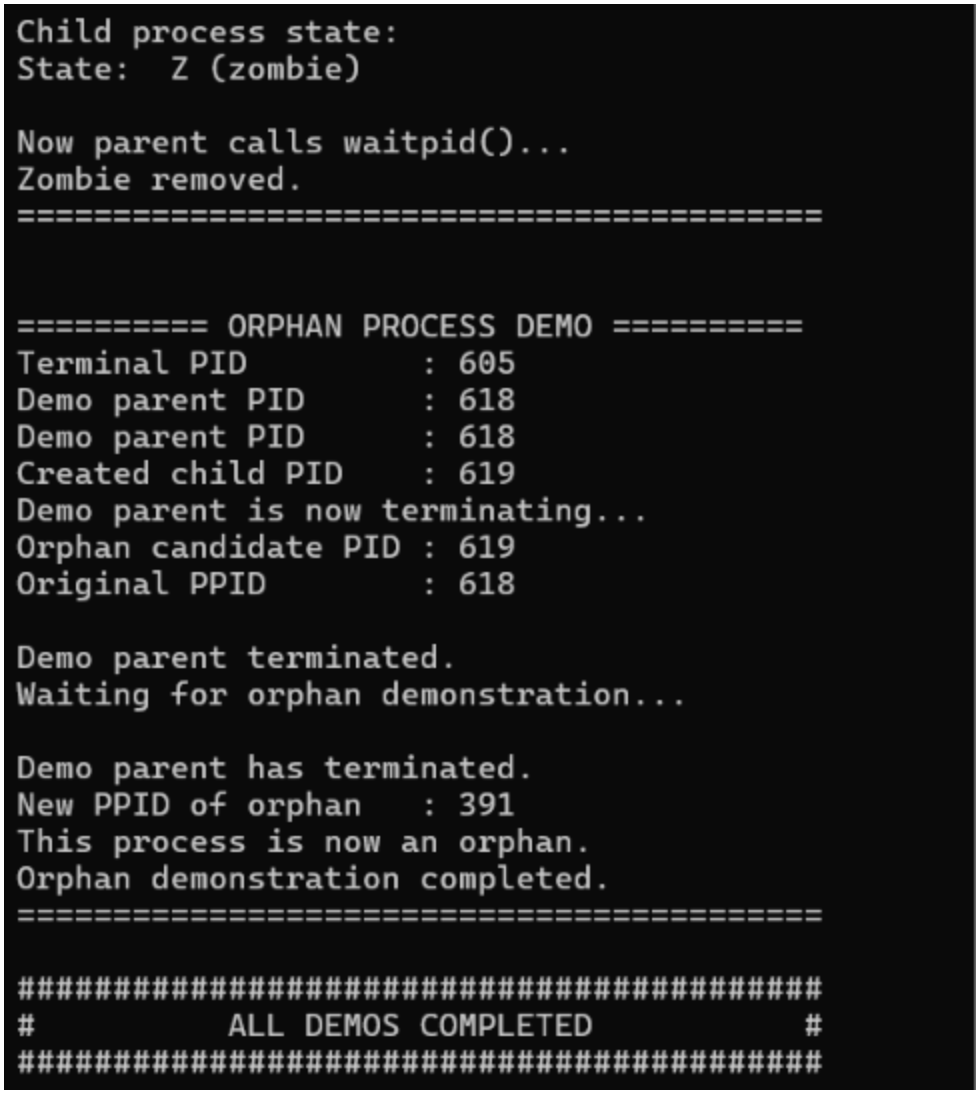
total 184
-rwxr-xr-x 1 geya geya 25904 Sep 22 06:57 project
-rw-r--r-- 1 geya geya 18296 Sep 24 10:00 project.c
-rwxr-xr-x 1 geya geya 25904 Oct  1 10:02 project1
-rw-r--r-- 1 geya geya 18324 Oct  1 10:02 project1.c
-rw-r--r-- 1 geya geya    25 Oct  1 10:02 project1.c:Zone.Identifier
-rwxr-xr-x 1 geya geya 25944 Oct  1 10:06 project2
-rw-r--r-- 1 geya geya 18820 Oct  1 10:06 project2.c
-rw-r--r-- 1 geya geya    25 Oct  1 10:06 project2.c:Zone.Identifier
-rwxr-xr-x 1 geya geya 25976 Oct  1 09:58 project_1
-rw-r--r-- 1 geya geya    25 Oct  1 09:57 project_1.c:Zone.Identifier
Child finished.
======================================
```

```
========== PIPE DEMO ==========
First child PID: 614
Parent PID: 605
Waiting for both processes...
Second child PID: 615
10
Pipe execution completed.
==============================


========== TERMINATION DEMO ==========
Parent PID: 605
Waiting for child...
Child PID: 616
Child is terminating using exit(5)
Child terminated normally.
Child exit status: 5
======================================


========== ZOMBIE PROCESS DEMO ==========
Parent PID: 605
Child PID : 617

Child has terminated.
Parent has NOT called wait() yet.
Checking child state from /proc...
Child PID: 617
Child is exiting now...
```


Child process state:
State: Z (zombie)
Now parent calls waitpid()...
Zombie removed.
==========
========== ORPHAN PROCESS DEMO ==========
Terminal PID : 605
Demo parent PID : 618
Demo parent PID : 618
Created child PID : 619
Demo parent is now terminating...
Orphan candidate PID : 619
Original PPID : 618
Demo parent terminated.
Waiting for orphan demonstration...
Demo parent has terminated.
New PPID of orphan : 391
This process is now an orphan.
Orphan demonstration completed.
==========
##########
ALL DEMOS COMPLETED
##########

Choice 7:

```
========================================
              MY OS TERMINAL
========================================
1. Fork + Exec Demo
2. Pipe Demo
3. Process Termination Demo
4. Zombie Process Demo
5. Orphan Process Demo
6. Run All OS Demos
7. Start Normal Terminal
8. Help
9. Exit
========================================
Enter your choice: 7

========================================
           MY OS TERMINAL
========================================
Type 'help' to see available commands.
Type 'exit' to return to the menu.
========================================

myterminal$
```

```
======================================
         MY OS TERMINAL
======================================
Type 'help' to see available commands.
Type 'exit' to return to the menu.
======================================

myterminal$ ls
project    project1    project1.c:Zone.Identifier  project2.c                project_1
project.c  project1.c  project2                    project2.c:Zone.Identifier  project_1.c:Zone.Identifier
myterminal$ whoami
geya
myterminal$ pwd
/home/geya/OS/Project
myterminal$
```

```
myterminal$ pipedemo

========== PIPE DEMO ==========
First child PID: 660
Parent PID: 605
Waiting for both processes...
Second child PID: 661
10
Pipe execution completed.
================================

Type 'exit' to return to the main menu.

myterminal$ zombiedemo

========== ZOMBIE PROCESS DEMO ==========
Parent PID: 605
Child PID : 662

Child has terminated.
Child PID: 662
Parent has NOT called wait() yet.
Checking child state from /proc...
Child is exiting now...

Child process state:
State:  Z (zombie)

Now parent calls waitpid()...
Zombie removed.
```

Type exit to go back to main menu from custom shell:

```
==========================================

Type 'exit' to return to the main menu.

myterminal$ exit
Returning to main menu...

========================================
             MY OS TERMINAL
========================================
1. Fork + Exec Demo
2. Pipe Demo
3. Process Termination Demo
4. Zombie Process Demo
5. Orphan Process Demo
6. Run All OS Demos
7. Start Normal Terminal
8. Help
9. Exit
========================================
Enter your choice:
```